DOROTEA MONACO

Aspiring AI/ML Engineer, Software Engineer

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PROFESSIONAL PROFILE

Master's student in Software Engineering at Politecnico of Turin with interests in AI, cloud technologies, and software development. Experienced in developing assistive solutions through volunteer work with Associazione Italiana Sclerosi Multipla, focusing on creating practical and human-centered technologies. Eager to apply technical skills to projects that combine innovation and social impact.

EDUCATION

09/2024
Turin, 28.88 GPA

MSc Computer Engineering - Software
Politecnico di Torino
- Advanced training in software system design, architecture, and development
- Coursework in machine learning, computer vision, large language models, and data science
- Experience with system programming, distributed systems, databases, and network technologies
- Focus on secure, scalable, and intelligent software systems

09/2020 – 03/2025
Turin, 98/110

BSc Computer Engineering
Politecnico di Torino
- Algorithms and Data Structures
- Programming Techniques
- Computer Architecture
- Databases
- Operating Systems
- Computer Networks

WORK EXPERIENCE

03/2024 – 07/2024
Turin

Comune di Torino, Circoscrizione 8
Internship
- Contributed to the digitization of the municipal archive •Supported website maintenance
- Provided digital support
- Created digital content to improve communication and accessibility of public information

SKILLS

Soft Skills
- Teamwork
- Problem Solving
- Organization & Time Management
- Resilience & Adaptability
- Continuous Learning

Technical Skills
- Programming Languages: Python, Java, C, Rust, JavaScript, TypeScript, SQL, Assembly
- Web Development: HTML, CSS, WebSockets
- Databases: PostgreSQL, MySQL, Redis
- Software Engineering & Systems: Software Architecture, System Programming, Formal Languages & Compilers, Information Systems
- Machine Learning & AI: Machine Learning, Computer Vision, Large Language Models for Software Engineering, LangChain
- Networking & Security
- Tools & Technologies: Git, GitHub, Linux, REST APIs

LANGUAGES

English
Professional working proficiency. Able to effectively conduct research and collaborate in English, supported by extensive experience developing academic and technical projects in English.

Italian
Native

PROJECTS

12/2025 – 02/2026

GAN for Data Augmentation and Domain Adaptation
Technologies: Python, PyTorch/TensorFlow, Deep Learning, GANs, Data Augmentation
Developed a deep learning pipeline using Generative Adversarial Networks (DCGAN and cDCGAN) for synthetic data generation and domain adaptation in imbalanced medical imaging datasets. Experimental results showed significant improvements in downstream classification performance, including up to +23.66% increase in recall and +0.16 improvement in F1-score for models trained from scratch, as well as overall accuracy gains up to ~90% on augmented datasets . Additionally, synthetic augmentation reduced class imbalance from 7.5:1 to 2.39:1, enabling more robust detection of rare malignant cases. This approach has strong real-world implications in medical diagnostics, where improving recall directly reduces false negatives, supporting earlier and more reliable disease detection in data-scarce clinical settings.

12/2025 – 01/2026

Architectures for Code Development with LLMs
Technologies: Python, Large Language Models (LLMs), AI-assisted software development
Designed and evaluated software architectures integrating Large Language Models (LLMs) into the code development workflow, comparing Naive, Single-Agent, and Multi-Agent approaches across 25 programming tasks spanning multiple domains . Experimental results showed that Multi-Agent systems achieved up to +10.8 percentage points improvement in logic tasks and +9.4 points in data structures and algorithms, reaching an overall correctness of 77.6%, while Single-Agent pipelines achieved 74.6%. Additionally, agent-based approaches improved code maintainability (Maintainability Index ≈ 61 vs. 59 baseline) despite increased complexity, demonstrating more robust handling of edge cases. These findings highlight practical benefits for real-world software engineering, where multi-agent LLM systems can enhance reliability and correctness in complex development tasks, while simpler architectures remain more efficient for routine coding, enabling adaptive AI-assisted development workflows in industry.

10/2025 – 01/2026

Participium

Technologies: React, TypeScript, Node.js, Express, PostgreSQL, Docker, RESTful APIs, HTML5, CSS3, Git, Jest (for testing)

Developed a full-stack web application enabling citizens to report urban issues (e.g., potholes, broken streetlights, waste) directly to municipal offices via an intuitive map-based interface. The platform features role-based access control, interactive geolocation, real-time status tracking, secure authentication, responsive UI, and automated tests. The project was executed using Scrum methodology, demonstrating skills in full-stack development, API design, containerized deployment, and collaborative software engineering in an academic team project.

07/2025 – 09/2025

Ruggine

Technologies: Rust, WebSocket, Tokio (async runtime), Iced (desktop GUI), SQLite/PostgreSQL, Redis, Docker, Git, AES-256-GCM encryption.

Developed a modern, secure, and scalable real-time chat application built entirely in Rust with a cross-platform desktop client. The system supports end-to-end encrypted messaging, WebSocket-based real-time communication, private and group chats, persistent message history, online presence tracking, and efficient asynchronous server architecture.

OPEN SOURCE

02/2026

LLM4S

Contributed multiple improvements and features to the llm4s open-source project, including:

Fixed broken and missing documentation links to improve usability and onboarding.

Added unit tests for ChunkerFactory, DocumentChunk, and SimpleChunk modules to ensure code reliability.

Implemented safe embedding validation in embedBatch to enhance robustness and prevent runtime errors.

Developed the AdaptiveWindowing pruning strategy with intelligent context window autotuning, including detailed documentation and production-ready examples.

These contributions demonstrate skills in Scala development, testing, documentation, realtime data processing, and collaborative open-source workflow using GitHub Pull Requests

02/2026

MoFA

Contributed multiple enhancements and fixes through merged pull requests, including:

Improved build infrastructure and CI support to streamline development workflows.

Refined command utilities and simplified configuration parsing for better developer experience.

Fixed critical bugs, strengthened type safety, and resolved edge-case panics to improve stability.

Enhanced feature support in core modules, expanding the agent runtime’s capability set.

Added comprehensive tests and validations to increase code quality and regression safeguards.

Upgraded documentation and examples to help new contributors understand core abstractions.

These contributions demonstrate expertise in systems programming (Rust), open-source collaboration, distributed development workflows, maintainability improvements, and quality-driven software engineering through community review and iterative refinement.

VOLUNTEER WORK

Associazione Italiana Sclerosi Multipla, Sezione di Torino

Volunteer, Member

During my volunteering experience, I was actively involved in the organization of informative events, fundraising activities, and awareness campaigns in companies and schools

My role included coordinating logistics, supporting communication with participants, and helping promote the organization's mission to different audiences